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Chapter 2 Hydrostatics

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Chapter 2 Hydrostatics (6 credit hours)

Learning points in this chapter:

1. Properties of Stress in Static Fluids
2. Balance equations; isobaric surface
3. Pressure profiles in stationary liquids and relative stationary liquids; absolute pressure, relative pressure, vacuum degree; pressure gauge head
4. Total liquid force acting on a plane; pressure center; pressure profile map method
5. Total liquid force acting on a surface; pressure body
6. Buoyancy; float balance

Chapter 2 Hydrostatics

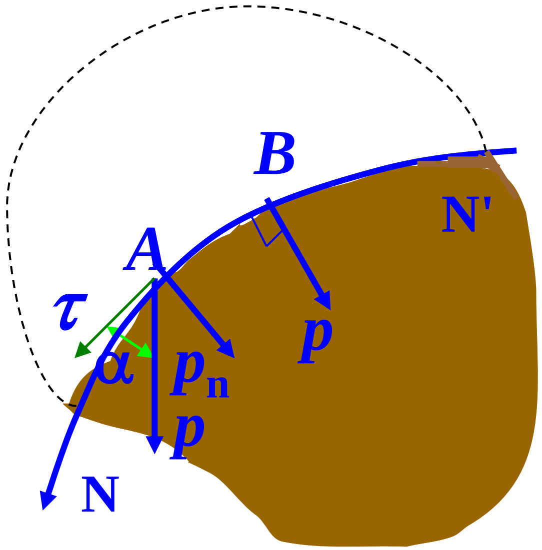
Section 1 Hydrostatic pressure characteristics

- Properties of stress at any point in a fluid at rest:

1, Pressure is directed negative to the normal of a surface

Fluid cannot withstand shear and is easy to flow (there is no hydrostatic shear stress)

2, Pressure is isotropic (equal in all directions)



$$p_x = p_y = p_z = p$$

However, the pressures at different points in a static fluid are generally unequal.

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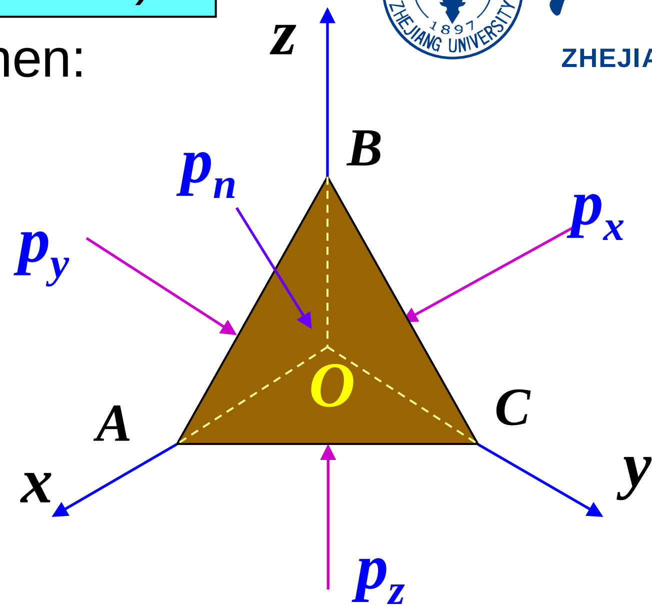
2, Pressure is isotropic (equal in all directions)

Proof: Take a tetrahedron ***OABC***, then:

$$\left.\begin{aligned} P_x - P_n\cos(n,x) + F_x &= 0 \\ P_y - P_n\cos(n,y) + F_y &= 0 \\ P_z - P_n\cos(n,z) + F_z &= 0 \end{aligned}\right\}$$



Force analysis in the **x**-direction:

Surface force: $\begin{cases} P_x = p_x \cdot \dfrac{1}{2}\mathrm{d}y\mathrm{d}z \\ P_n\cos(n,x) = p_n \cdot \dfrac{1}{2}\mathrm{d}y\mathrm{d}z \end{cases}$

n is the normal direction of the inclined plane ***ABC***

Mass force: $F_x = f_x \cdot \rho\mathrm{d}x\mathrm{d}y\mathrm{d}z/6$

$$p_x - p_n + \frac{1}{3}\mathrm{d}x \cdot \rho f_x = 0$$

When the tetrahedron becomes infinitely small, then **d*x*→0**, so

$$p_x = p_n$$

Same for x and y directions:

$$p_x = p_y = p_z = p_n$$

Hence, orientation of the surface does not matter.

This is

known a`Pascal's law

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Side remark:

(1) The actual fluid in the motion state produces shear stress, then the pressure components are no longer equal and the pressure is given by

$$p = \frac{1}{3}(p_x + p_y + p_z)$$

(2) However, when the moving fluid is an ideal fluid, there will be no shear stress, and the pressure is given by

$$p_x = p_y = p_z = p$$

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Chapter 2 Hydrostatics

Section 1 Hydrostatic pressure characteristics

Section 2 Stress equilibrium

Section 3 Pressure profile of a static fluid

Section 4 Manometer

Section 5 Hydrostatic pressure on a plane

Section 6 Hydrostatic pressure on a surface

Section 7 Buoyancy and floating-body stability

Chapter summary

Chapter 2 Hydrostatics

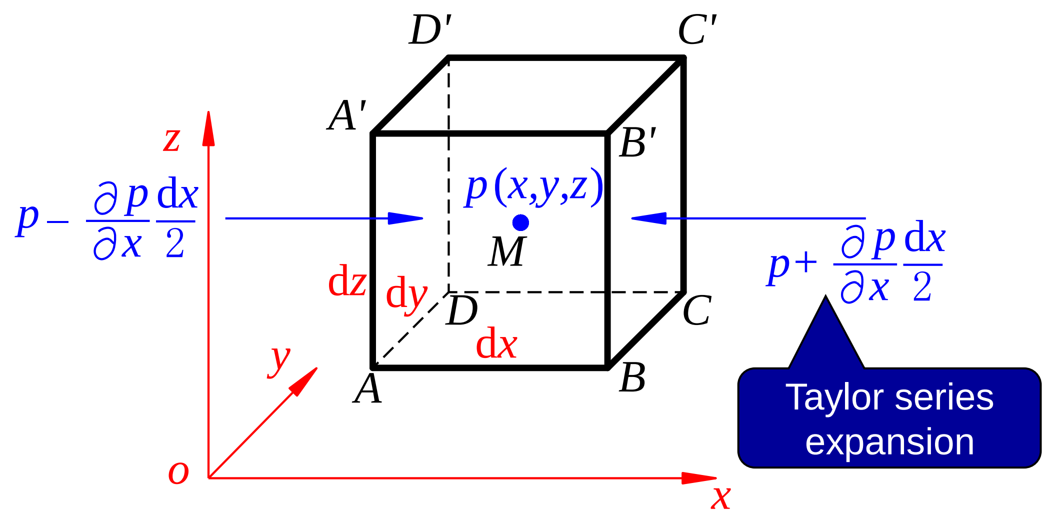
Section 2 Stress equilibrium

Section 2 Balance equations

1. Stress equilibrium

Force in ***x***-direction:

Surface force $\begin{cases} (p-\dfrac{\partial p}{\partial x}\dfrac{\mathrm{d}x}{2})\mathrm{d}y\mathrm{d}z \\ \\ (p+\dfrac{\partial p}{\partial x}\dfrac{\mathrm{d}x}{2})\mathrm{d}y\mathrm{d}z \end{cases}$



Mass force: $f_x \rho \mathrm{d}x\mathrm{d}y\mathrm{d}z$

$\sum F_x = 0$, then:

$$(p-\frac{\partial p}{\partial x}\frac{\mathrm{d}x}{2})\mathrm{d}y\mathrm{d}z-(p+\frac{\partial p}{\partial x}\frac{\mathrm{d}x}{2})\mathrm{d}y\mathrm{d}z+f_x\rho\mathrm{d}x\mathrm{d}y\mathrm{d}z=0$$

Rearranged: $f_x-\dfrac{1}{\rho}\dfrac{\partial p}{\partial x}=0$

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Stress equilibrium:

$$\left.\begin{aligned} f_x - \frac{1}{\rho}\frac{\partial p}{\partial x} = 0 \\ f_y - \frac{1}{\rho}\frac{\partial p}{\partial y} = 0 \\ f_z - \frac{1}{\rho}\frac{\partial p}{\partial z} = 0 \end{aligned}\right\} \qquad (1)$$

Physical meaning: In a fluid in equilibrium, the surface force component and mass force component of a unit mass of fluid compensate each other. The rate of change of pressure along the axial direction ($\frac{\partial p}{\partial x}, \frac{\partial p}{\partial y}, \frac{\partial p}{\partial z}$) is equal to the mass force per unit volume in that axial direction (ρf_x, ρf_y, ρf_z).

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Chapter 2 Hydrostatics

Section 2 Stress equilibrium

1, Stress equilibrium

2, Consequences of stress equilibrium

3, Isobaric surface

2, Consequences of stress equilibrium

$$\left.\begin{aligned} f_x - \frac{1}{\rho}\frac{\partial p}{\partial x} &= 0 \\ f_y - \frac{1}{\rho}\frac{\partial p}{\partial y} &= 0 \\ f_z - \frac{1}{\rho}\frac{\partial p}{\partial z} &= 0 \end{aligned}\right\} \quad (1)$$

p = p(x,y,z)

Pressure differential: $\mathrm{d}p = \frac{\partial p}{\partial x}\mathrm{d}x + \frac{\partial p}{\partial y}\mathrm{d}y + \frac{\partial p}{\partial z}\mathrm{d}z$

The terms of (1) are multiplied by dx, dy, and dz and added together to get

$$f_x\mathrm{d}x + f_y\mathrm{d}y + f_z\mathrm{d}z = \frac{1}{\rho}\left(\frac{\partial p}{\partial x}\mathrm{d}x + \frac{\partial p}{\partial y}\mathrm{d}y + \frac{\partial p}{\partial z}\mathrm{d}z\right)$$

$$\therefore \mathrm{d}p = \rho(f_x\mathrm{d}x + f_y\mathrm{d}y + f_z\mathrm{d}z)$$

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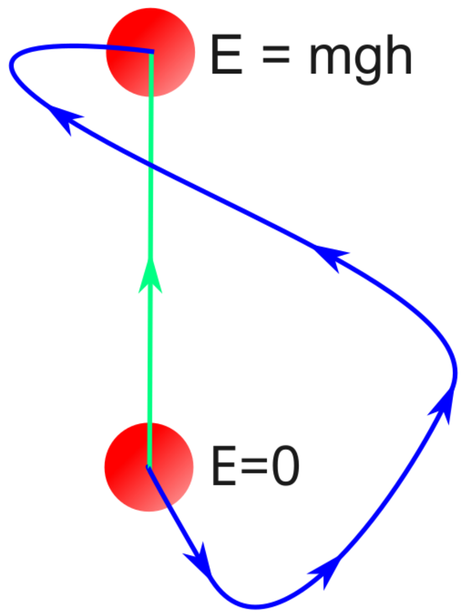
A conservative force exists when the work done by that force on an object is independent of the object's path. Instead, the work done by a conservative force depends only on the end points of the motion. An example of a conservative force is gravity.



Physical meaning of $\mathrm{d}p = \rho(f_x \mathrm{d}x + f_y \mathrm{d}y + f_z \mathrm{d}z)$

If ρ **= const**， defined ***p*** **/**ρ **=*w*(*x*,*y*,*z*)** , then:

$$\mathrm{d}\left(\frac{p}{\rho}\right) = \mathrm{d}(w) = f_x \mathrm{d}x + f_y \mathrm{d}y + f_z \mathrm{d}z = \frac{\partial w}{\partial x}\mathrm{d}x + \frac{\partial w}{\partial y}\mathrm{d}y + \frac{\partial w}{\partial z}\mathrm{d}z$$

Shows that ***w*** is a potential function, and the force ***f*** is a conservative force.

Conclusions:

a. Constant density fluids can maintain stress equilibrium only under the action of conservative mass forces

b. The gravity and inertial forces are conservative forces

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Chapter 2 Hydrostatics

Section 2 Stress equilibrium

1, Stress equilibrium

2, Consequences of stress equilibrium

3, Isobaric surface

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3, Isobaric surface

Isobaric surface: the surface formed by points of equal pressure (**p=C**).

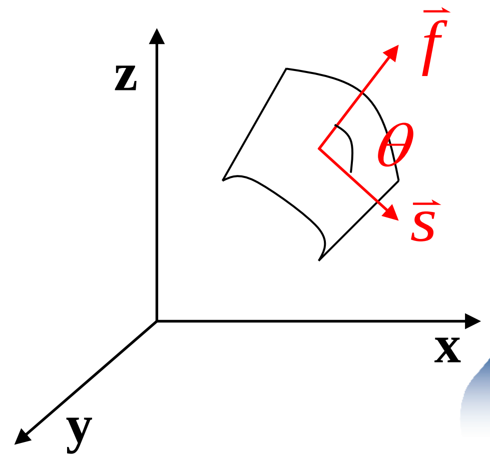
Common isobaric surfaces are: the free liquid surface and the interface between two fluids that do not mix with each other in a fluid in equilibrium.

$$\because\ p = C,\quad \mathrm{d}p = 0$$

$$\therefore \frac{\mathrm{d}p}{\rho} = f_x\mathrm{d}x + f_y\mathrm{d}y + f_z\mathrm{d}z = 0$$

Important property: the mass force at any point of the isobaric surface of a fluid in equilibrium is always orthogonal to the isobaric surface.

$$\vec{f}\cdot\mathrm{d}\vec{s} = 0$$



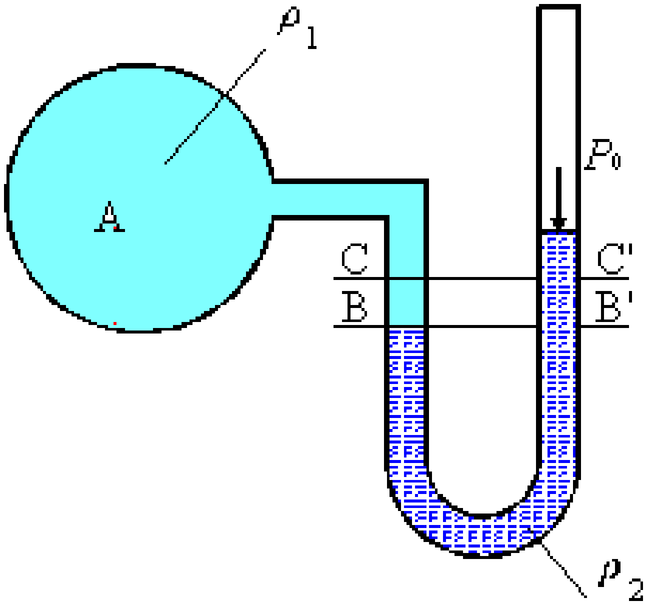
Proof: $\vec{f} = f_x\vec{i} + f_y\vec{j} + f_z\vec{k}$

$$\mathrm{d}\vec{s} = \mathrm{d}x\vec{i} + \mathrm{d}y\vec{j} + \mathrm{d}z\vec{k}$$

$$\vec{f}\cdot\overrightarrow{\mathrm{d}s} = f_x\mathrm{d}x + f_y\mathrm{d}y + f_z\mathrm{d}z = 0$$

An isobaric surface under the action of gravity should meet the conditions:

1. Standing still (Fluid is at rest);
2. Connected (fluid region is continuous);
3. The connecting medium is the same homogeneous fluid;
4. The force of mass is only gravity;
5. The same level.



In the picture on the right, the B-B' plane is the isobaric plane.

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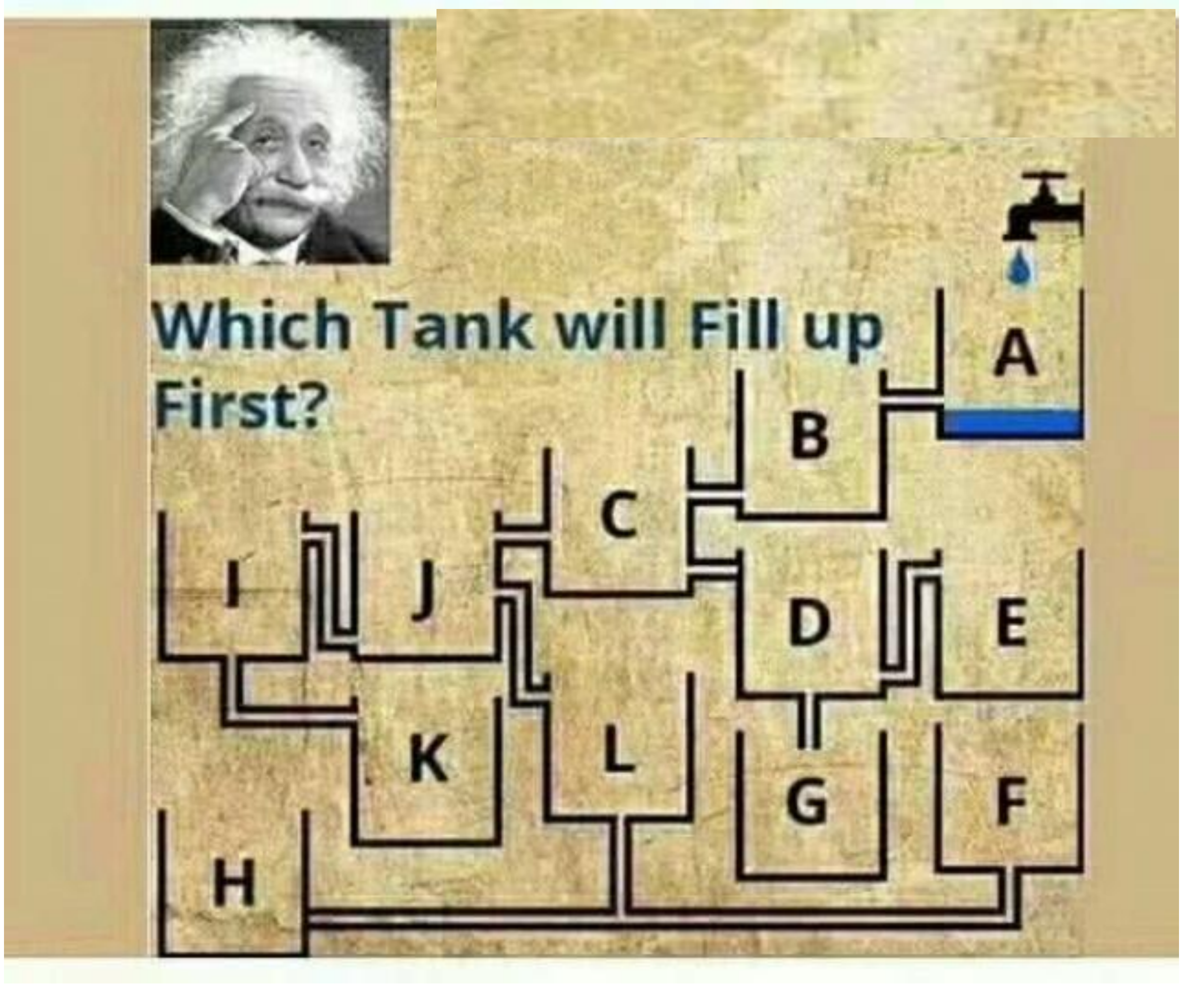
Question:



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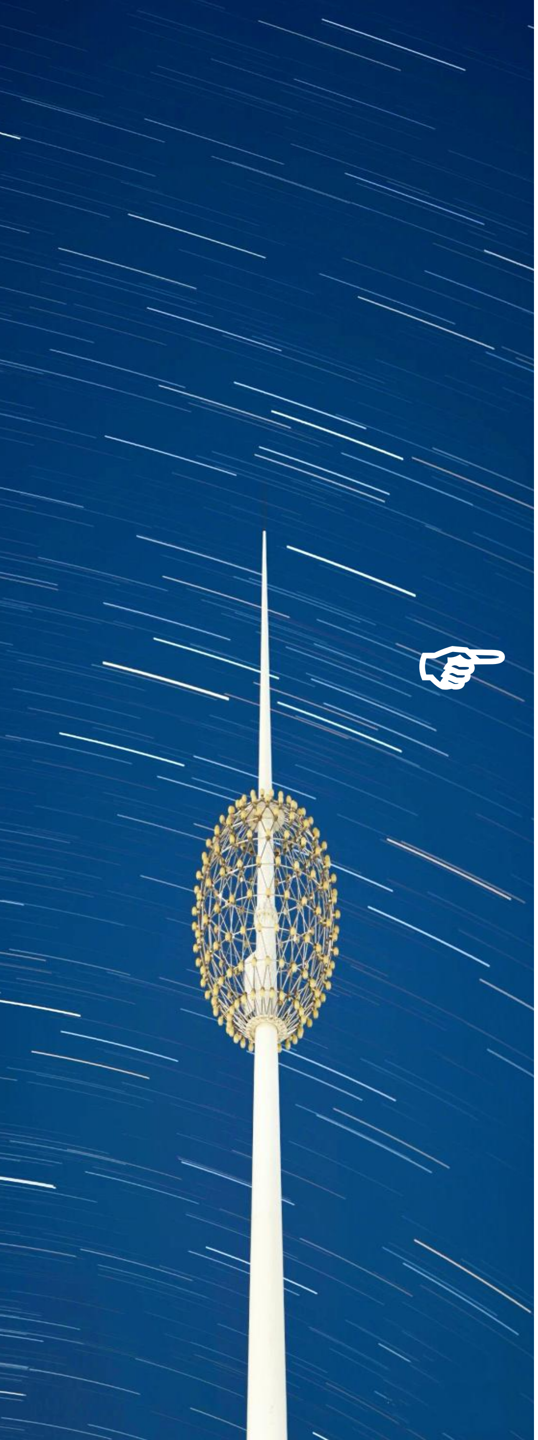
Chapter 2 Hydrostatics



Chapter 2 Hydrostatics

Section 3 Pressure profile of a static fluid

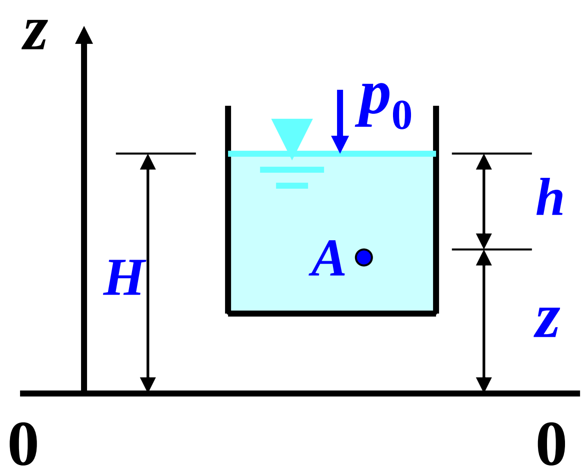
1, Pressure profile under gravity

2, Pressure representation

3, Pressure profile at relative equilibrium

Section 3 Pressure profile of a static fluid

1, Pressure profile under gravity



Force per unit mass:

$$f_x = f_y = 0,\ \ f_z = -g$$

Substitute into stress equilibrium equation

$$\mathrm{d}p = \rho(f_x \mathrm{d}x + f_y \mathrm{d}y + f_z \mathrm{d}z)$$

$$\mathrm{d}p = -\rho g \mathrm{d}z$$

$$p = -\rho g z + C$$

At free surface: $z = H,\ \ p = p_0$

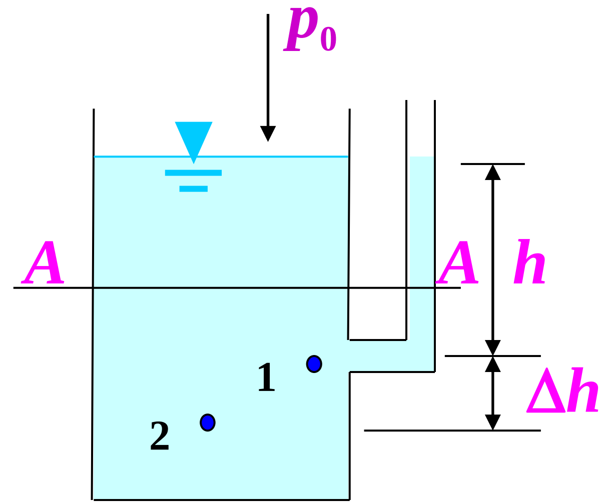
$$C = p_0 + \rho g H$$

Hydrostatic pressure:

$$p = p_0 + \rho g (H - z) = p_0 + \rho g h$$

Hydrostatic pressure:

$$p = p_0 + \rho g h$$



Conclusion for a homogeneous fluid under gravity:

1) The hydrostatic pressure increases linearly with depth.

2) The hydrostatic pressure is equal to the product of the surface pressure plus the ρg of the fluid and the submerged depth at that point.

3) The pressure at each point with the same depth ***h*** under the free surface is equal. The isobaric surface of a continuously connected static fluid is the horizontal surface.

4) Extension: Knowing the pressure at a point and the depth difference between two points, the pressure value at another point can be obtained.

$$p_2 = p_1 + \rho g \Delta h$$

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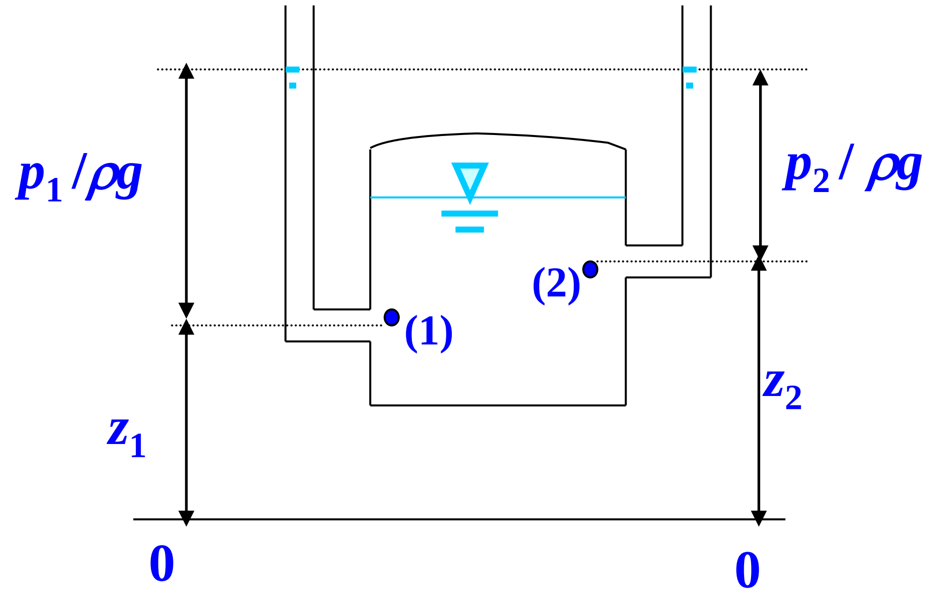
- Energy balance of a static fluid

  Hydrostatic pressure：

$$\frac{p}{\rho g} + z = C$$

or

$$z_1 + \frac{p_1}{\rho g} = z_2 + \frac{p_2}{\rho g}$$



Elevation head z: the height of any point above the reference plane **0-0**. It represents the positional potential energy of a unit weight of fluid.

Pressure head ***p/ρg***: the pressure potential energy per unit weight of fluid.

Hydraulic head ($z + \frac{p}{\rho g}$): the total potential energy per unit weight of fluid.

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Conclusion - Different expressions of the pressure profile under gravity in a stationary fluid:

1, $p = p_0 + \rho g h$, pressure increases linearly with depth;

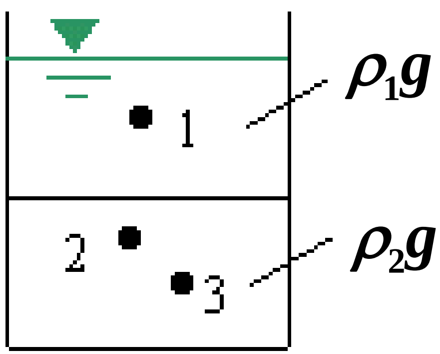
2, $z + \dfrac{p}{\rho g} = C$, hydraulic head is constant;

3, Total potential energy per unit weight is constant;

4, Water surface is the reference elevation for the pressure head

5, The isobaric surface is a horizontal surface.

In the figure, which of the two equations is correct, $\rho_1 g \neq \rho_2 g$?



$$z_1 + \frac{p_1}{\rho g} = z_2 + \frac{p_2}{\rho g}$$

$$z_3 + \frac{p_3}{\rho g} = z_2 + \frac{p_2}{\rho g}$$

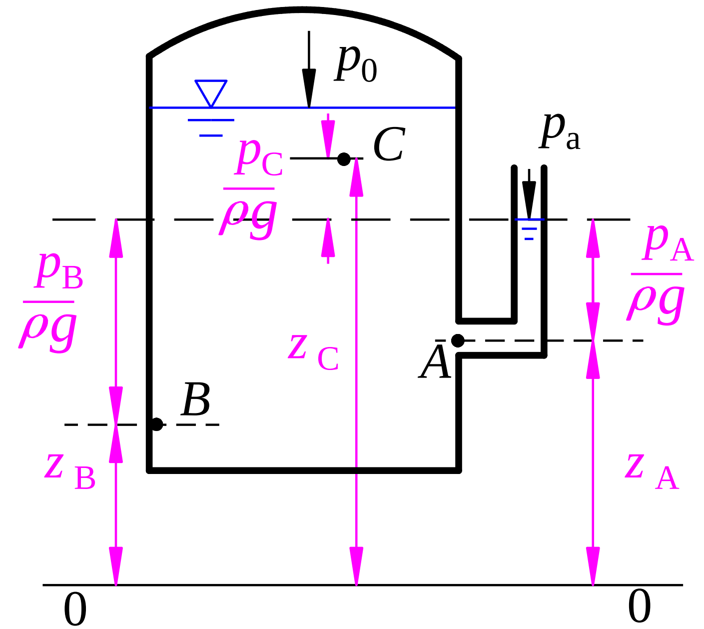
Example: Try to mark the elevation head, pressure head, and hydraulic head of three points ***A***, ***B***, and ***C*** in the liquid container, as shown in the figure. Take the picture **0-0** as the reference plane.

A: pressure head $\frac{p_A}{\rho g}$, elevation head z_A, hydraulic head $z_A + \frac{p_A}{\rho g}$

B: analogous

C: relative pressure is negative, $p_C < p_a$.

Therefore, negative pressure head $\frac{p_c}{\rho g}$

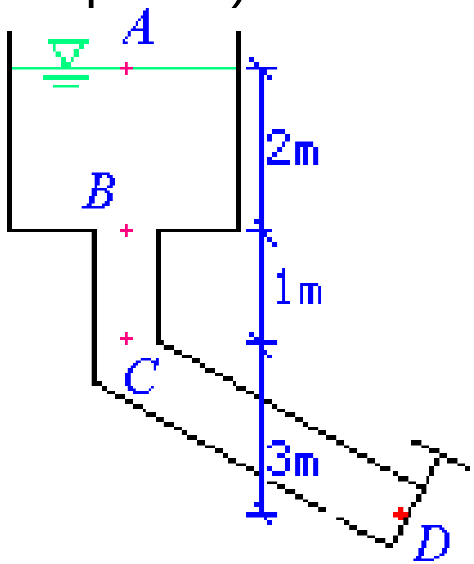


Question 1: Under only gravity, what is the total potential energy of the fluid per unit weight at any point in the stationary liquid to the same reference plane?

A. increases with depth;
B. constant;
C. decreases with depth;
D. uncertain.

Question 2: What are the pressure head and hydraulic head at points A, B, C, and D in the figure. (The gate at point D is closed, and the horizontal plane where point D is located is the reference plane.)

	Pressure head	Hydraulic head
A	**0m**	**6m**
B	**2m**	**6m**
C	**3m**	**6m**
D	**6m**	**6m**



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Chapter 2 Hydrostatics

Section 3 Pressure profile of a static fluid

1, Pressure profile under gravity

2, Pressure representation

3, Pressure profile at relative equilibrium

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2, Pressure representation

1, Expression of pressure

a. Absolute Pressure: $p_{abs} \geq 0$.

b. Relative Pressure: also known as "gauge pressure", $p = p_{abs} - p_a$.

It can be positive, negative or zero.

c. Vacuum: $p_{abs} < p_a = 1at$

Vacuum degree p_v

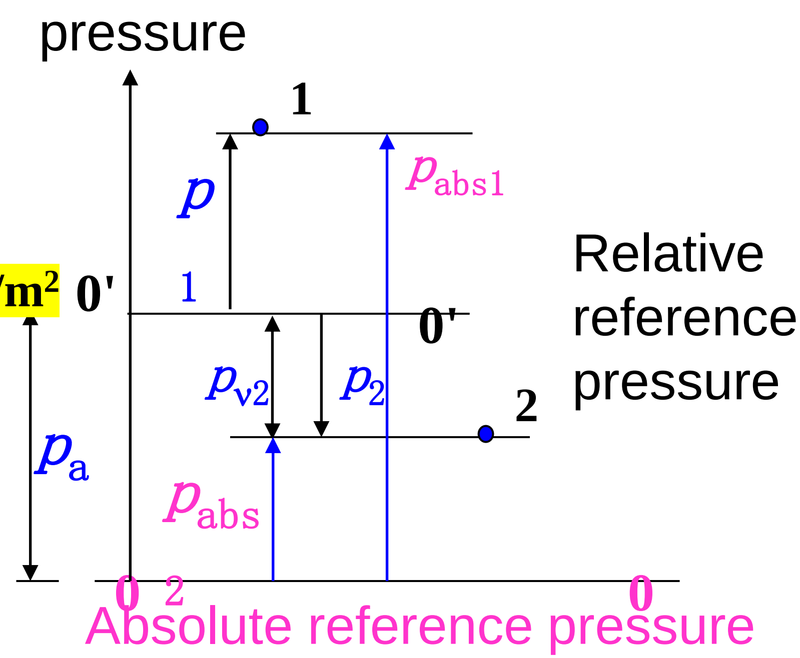
$$p_v = p_a - p_{abs} \qquad (p_{abs} < p_a)$$

1at=98kN/m²

vacuum height

$$h_v = \frac{p_v}{\rho g} = \frac{p_a - p_{abs}}{\rho g}$$

Note: Relative pressure is used for calculations unless otherwise specified.



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2, Unit of pressure

a. Stress unit

N/m^2, Pa

b. Atmospheric pressure

Standard atmospheric pressure: 1 standard atmospheric pressure (**atm**) = **1.013x10^5 Pa=101.3 kPa**

Engineering atmospheric pressure: 1 engineering atmospheric pressure (**at**) = **9.8x10^4 Pa=98 kPa**

c. Liquid column height

mH_2O: 1 mH_2O = 9.8 kPa

mHg: 1 mHg = 133 kPa

Example 1: Find the absolute and relative pressured at a depth of **2m** below the free surface of fresh water (the absolute pressure of the free surface is considered to be 1 engineering atmosphere)

Absolute pressure:

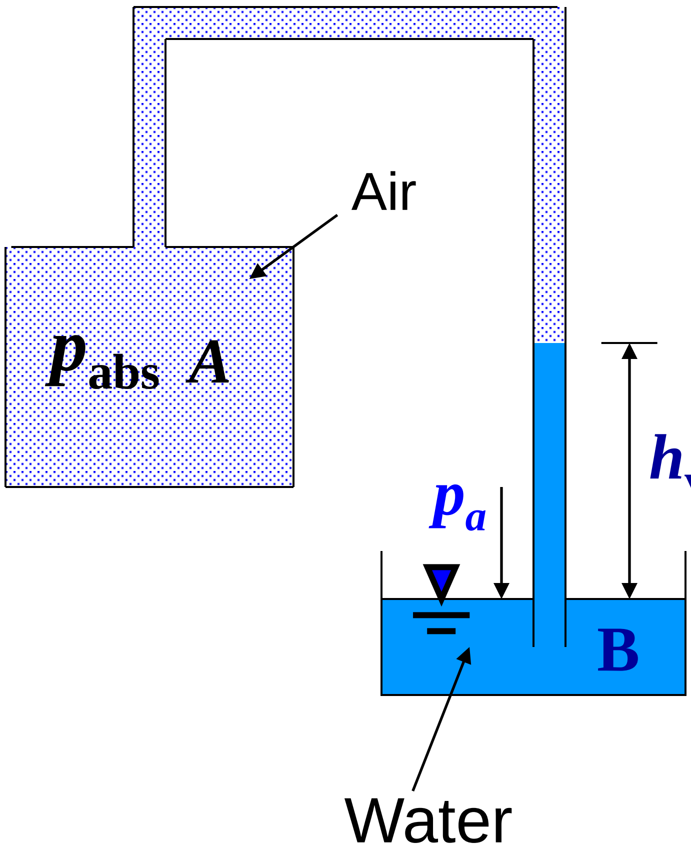
$$p_{\text{abs}} = p_0 + \rho g h = p_{\text{a}} + \rho g h = 98000\,\text{N/m}^2 + 9800\,\text{N/m}^3 \times 2\text{m}$$
$$= 117.6\ \text{kPa} = 1.2\ \text{at}$$

Relative pressure:

$$p = p_{\text{abs}} - p_{\text{a}} = \rho g h = 9800 \times 2 = 19.6\ \text{Pa} = 0.2\ \text{at}$$

Example 2 As shown in the figure, when $\boldsymbol{h_v}$**=2m**, find the vacuum degree in the closed container ***A***.

$$p_v = \rho g h_v = 9800 \times 2 = 19.6\ \text{kPa}$$



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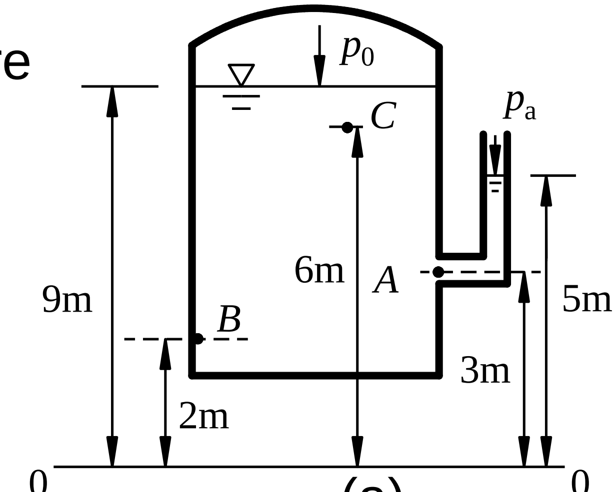
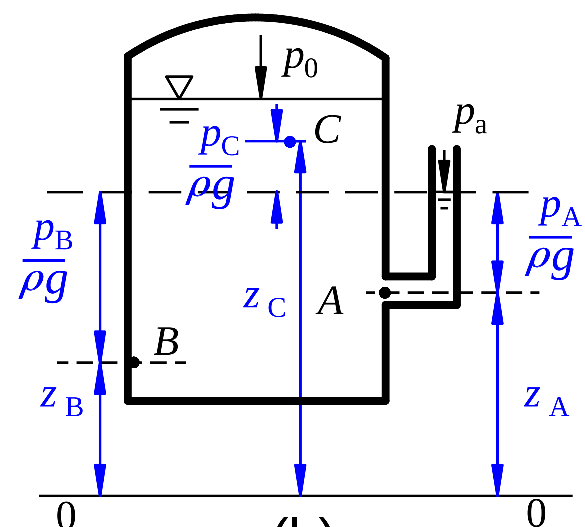
Example 3 Try to mark the elevation head, pressure head, and hydraulic head of the three points ***A***, ***B***, and ***C*** in the liquid container as shown in the figure (a).

Take **0–0** as the reference plane.



(a)



(b)

A: Elevation head = **3m**, pressure head = **2m**, hydraulic head = **5m**

B: Elevation head = **2m**, pressure head = **3m**, hydraulic head = **5m**

C: Elevation head = **6m**, pressure head = **-1m**, hydraulic head = **5m**

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Example 4 Determine the difference between the hydraulic heads at points ***A*** and ***B*** with a mercury U-shaped differential pressure gauge (see figure).

Solution: Define reference plane **0–0** and isobaric surface ***NM*** as in figure. Then:

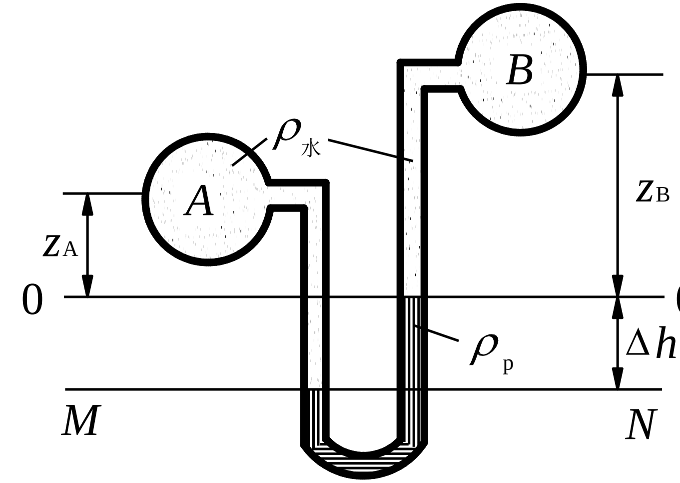
$$p_M = p_A + \rho g z_A + \rho g \Delta h_p$$

$$p_N = p_B + \rho g z_B + \rho_p g \Delta h_p$$

$$p_M = p_N$$

$$p_A - p_B = (\rho_p g - \rho g)\Delta h_p + \rho g (z_B - z_A)$$

$$(z_A + \frac{p_A}{\rho g}) - (z_B + \frac{p_B}{\rho g}) = (\frac{\rho_p - \rho}{\rho})\Delta h_p$$



Mercury-water-density ratio = 13.6 ->

$$(z_A + \frac{p_A}{\rho g}) - (z_B + \frac{p_B}{\rho g}) = 12.6\Delta h_p$$

Example 5 Vacuum degree of ***A*** is **17200 N/m²**. Each elevation is shown in the figure, ρg_1**=6860N/m³**, ρg_2**= 15680N/m³**. Try to find the elevation of the liquid level in the piezometer tube ***E***, ***F***, ***G*** and the height difference H_1 of the rising mercury in the U-shaped piezometer tube.

Solution: Use principle of isobaric surface

（1）E: $p_A+(\rho g)_1 h_1=p_a$ $\quad \therefore h_1=\dfrac{p_a-p_A}{(\rho g)_1}=2.5m$

Elevation $\nabla E=15.0-h_1=12.5\text{m}$

（2）F: $p_A+(\rho g)_1(15-11.6)=(\rho g)_{水}h_2+p_a$

$$h_2=\frac{p_A-pa+3.4(\rho g)_1}{(\rho g)_{水}}=0.62\text{m}$$

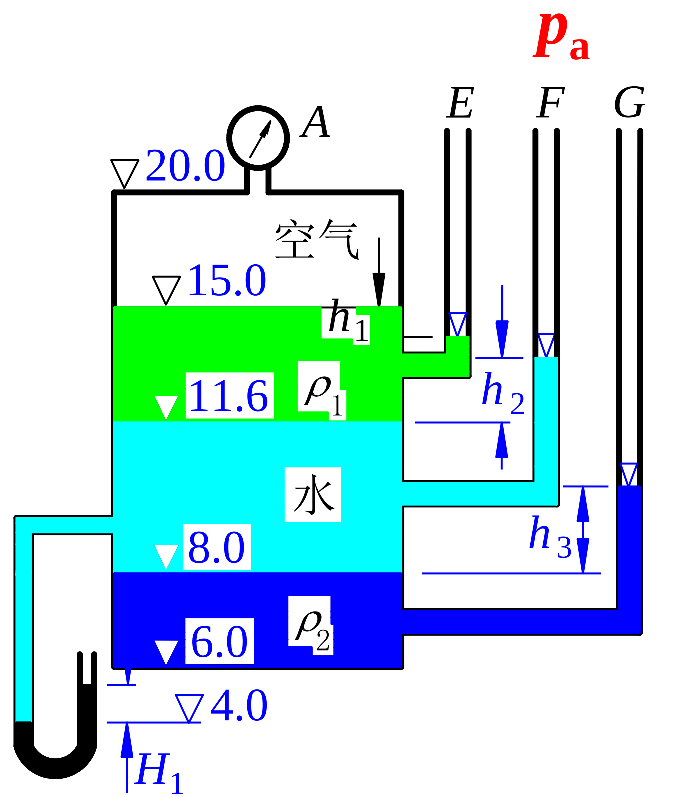
$\nabla F=\nabla 11.6+h_2=12.22\text{m}$

（3）G: $p_A+(\rho g)_1(15-11.6)+(\rho g)_{水}(11.6-8.0)$

$=(\rho g)_2 h_3+p_a$

$$h_3=\frac{p_A-pa+3.4(\rho g)_1+3.6(\rho g)_{\mathrm{w}}}{(\rho g)_{1_2}}=2.64\text{m}$$

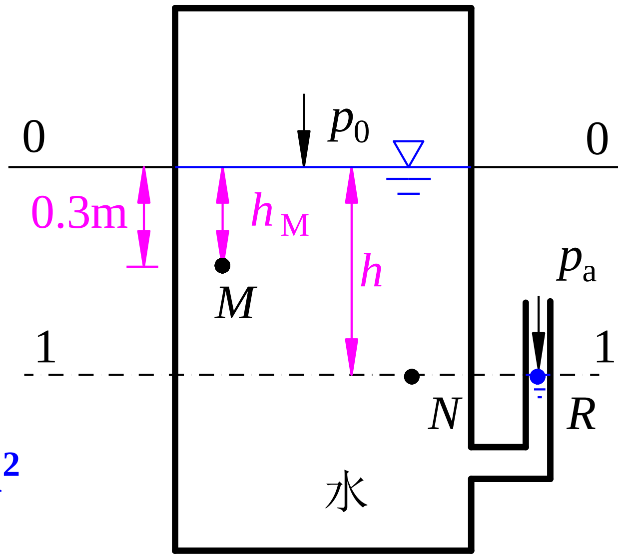
$\therefore \nabla G=\nabla 8.0+h_3=\nabla 10.64\text{m}$



（4）U: $p_A+(\rho g)_1(15-11.6)+(\rho g)_{水}(11.6-4.0)=(\rho g)_m H_1+p_a$

$$H_1=\frac{p_A-pa+3.4(\rho g)_1+7.6(\rho g)_{\mathrm{w}}}{(\rho g)_{\mathrm{m}}}=0.605\text{m}$$

Example 6 A sealed water tank is shown in the figure. If the relative pressure on the water surface is p_0=-44.5KN/m^2, find: (1) ***h*** value; (2) find the relative pressure at point ***M*** at **0.3m** underwater; (3) find the pressure head of point ***M*** relative to the reference plane **0-0**.



Solution: (1) ***h***

$$p_0+\rho g h=0,\quad h=44.5/9.8=4.54\ \text{m}$$

(2) p_M

Relative pressure:

$$p_M=p_0+\rho g h_M= -44.5+9.8\times 0.3= -41.56\ \text{kN/m}^2$$

$$p_M= -41.56/98= -0.424\text{at}\ (1\text{at}= 98\text{kN/m}^2)$$

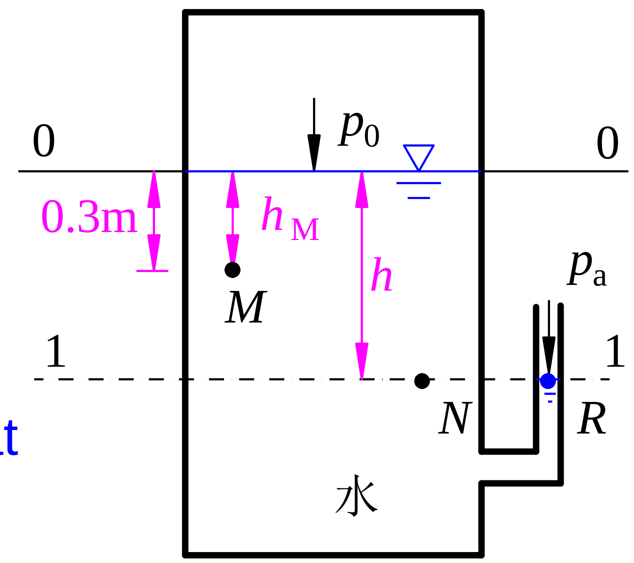
$$\frac{p_M}{\rho g}=\frac{-41.56}{9.8}=-4.24\text{m}$$

Absolute pressure:

$$p_{Mabs} = p_M + p_a = -41.56+98= 56.44\ \text{kN/m}^2$$

$$p_M = 56.44/98=0.576\ \text{at}$$

$$h_M = \frac{p_{Mabs}}{\rho g} = \frac{56.44}{9.8} = 5.76\text{m}$$



Vacuum degree: $p_v = 41.56\ \text{kN/m}^2 = 0.424\ \text{at}$

Vacuum height: $h_v = \frac{p_v}{\rho g} = \frac{41.56}{9.8} = 4.24\text{m}$

Question 1: The reading value of a pressure gauge is?

A. absolute pressure;

B. relative pressure;

C. Absolute pressure plus local atmospheric pressure;

D. Relative pressure plus local atmospheric pressure.

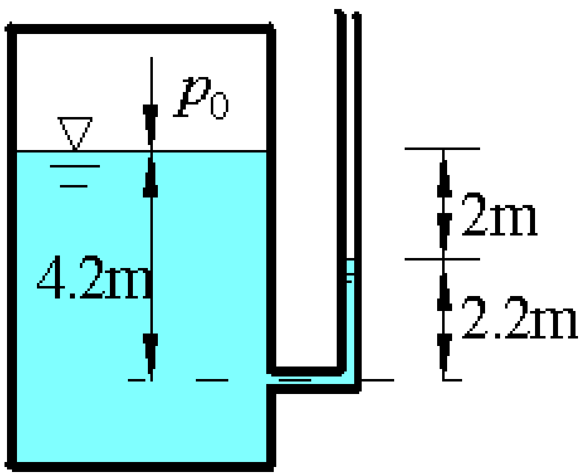
Question 2: The lower part of a closed container is water, the upper part is air at **1at** atmospheric pressure. What is the absolute pressure $\boldsymbol{p_0}$ of the liquid level in the container? How many meters of water column?

A, 2m；

B, 1m；

C, 8m；

D, -2m.



Chapter 2 Hydrostatics

Section 3 Pressure profile of a static fluid

1, Pressure profile under gravity

2, Pressure representation

☞ 3, Pressure profile at relative equilibrium

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3, Pressure profile at relative equilibrium

Relative equilibrium: refers to the relative static or relative equilibrium state without relative motion between the fluid particles and fluid-containing reference system. Because there is no relative motion between the fluid particles, there is no shear stress inside the fluid or between the fluid and the side walls.

In a fluid at relative equilibrium, in addition to gravity, the mass force is also affected by the inertial force.

1, Liquid in uniform acceleration motion

$$dp = \rho\,(f_x dx + f_y dy + f_z dz)$$

Mass force: $f_x = -a\cos\alpha$

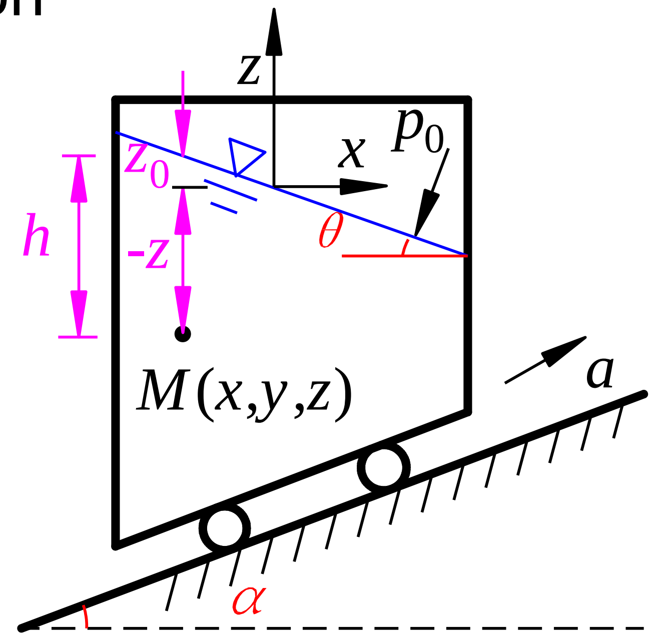
$f_y = 0$

$f_z = -(g + a\sin\alpha)$

$$p = -\rho\,[a\cos\alpha\, x + (g + a\sin\alpha) z] + C$$

$x = z = 0,\ p = p_0 \ \rightarrow\ C = p_0$

$$\therefore\ p = -\rho(g + a\sin\alpha)\left[\frac{a\cos\alpha}{g + a\sin\alpha}x + z\right] + p_0$$



Isobaric surface: $\dfrac{a\cos\alpha}{g + a\sin\alpha}x + z = const$

$$\theta = \tan^{-1}\left(-\frac{a\cos\alpha}{g + a\sin\alpha}\right)$$

Liquid level: $z_0 = -\dfrac{a\cos\alpha}{g + a\sin\alpha}x$

Pressure at ***M***: $p = p_0 + \rho(g + a\sin\alpha)\underbrace{(z_0 - z)}_{h}$ ——linear profile

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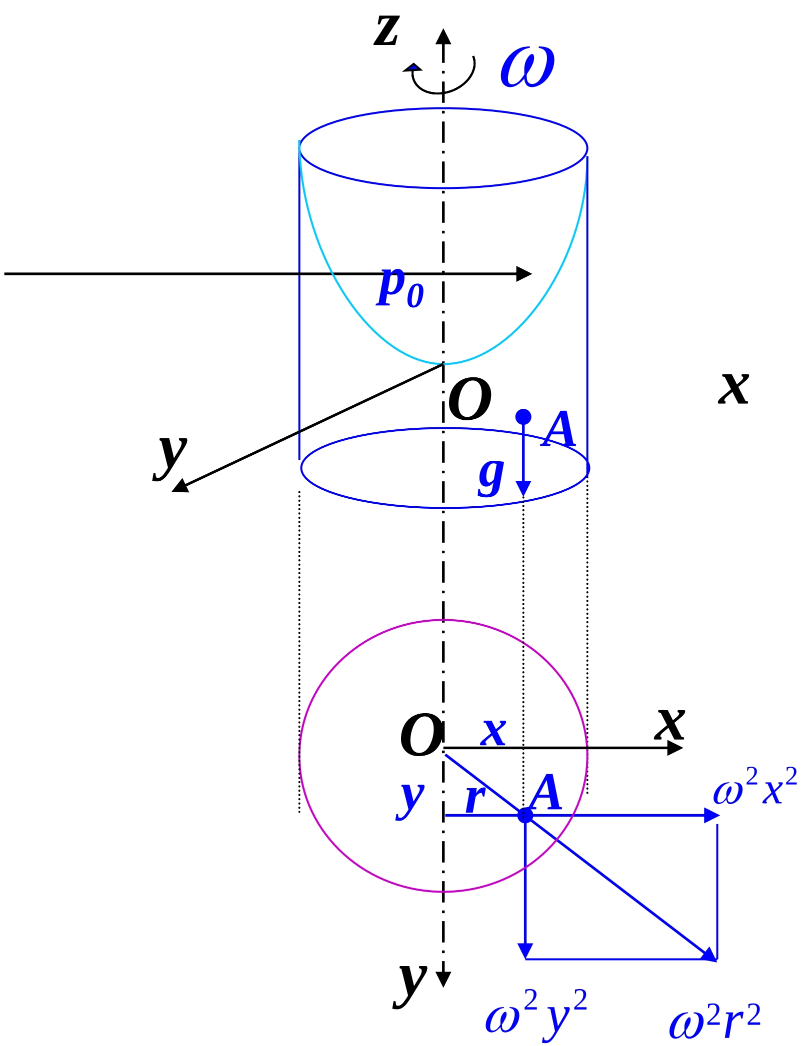
2, Rotating liquid

$$F = ma = m \cdot \frac{v^2}{r} = m \cdot \frac{(\omega r)^2}{r} = m\omega^2 r$$

$$f_x = \omega^2 x \quad f_y = \omega^2 y \quad f_z = -g$$

$$\therefore dp = \rho(\omega^2 x dx + \omega^2 y dy - g dz)$$

$$p = \rho\left(\frac{1}{2}\omega^2 x^2 + \frac{1}{2}\omega^2 y^2 - gz\right) + C$$



At origin (x=0, y=0, z=0): $p = p_0$

$$\therefore \quad C = p_0$$

$$\because \quad x^2 + y^2 = r^2$$

- General expression:

$$p = p_0 + \rho g\left(\frac{\omega^2 r^2}{2g} - z\right)$$

$$\boxed{p = p_0 + \rho g\left(\frac{\omega^2 r^2}{2g} - z\right)}$$

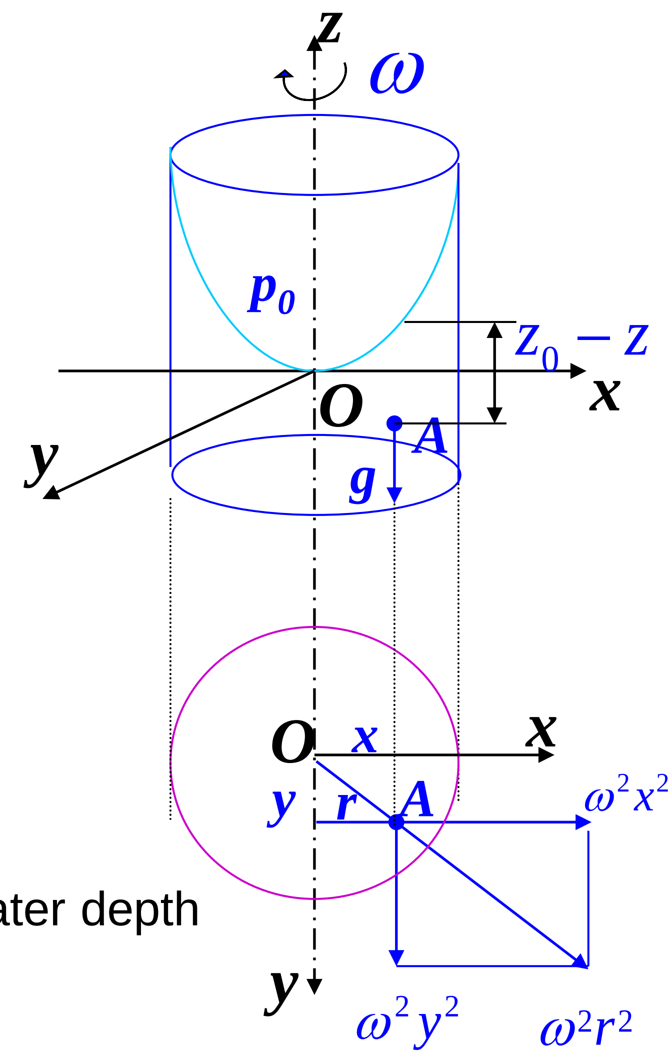
- Isobaric surface:

$$\frac{\omega^2 r^2}{2g} - z = C_1$$

- Free surface: $\because \quad p = p_a = p_0$

$$\therefore \quad z_0 = \frac{\omega^2 r^2}{2g}$$

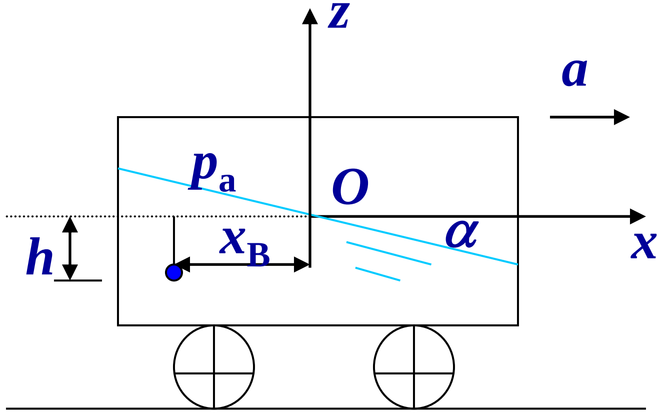
- Pressure: $p = p_0 + \rho g(z_0 - z) = p_0 + \rho g h$



Note 1: Pressure still has a linear relationship with water depth

Note 2: The hydraulic head is not constant

Example 1 A water truck drives horizontally forward with constant acceleration a**=0.98 m/s²**, find the angle α between the free surface and the horizontal plane; before motion point B is h **=1.0 m** below the water surface, and the distance from the z-axis is $x_B = -1.5$**m.** Find the hydrostatic pressure at this point after the truck accelerates.



Mass force: $f_x = -a;\ f_y = 0;\ f_z = -g$

$$dp = \rho(f_x dx + f_y dy + f_z dz)$$

$$dp = \rho(-adx - gdz)$$

$$p = -\rho(ax + gz) + C$$

$$x = z = 0;\quad p = p_0$$

$$C = p_0 = 0$$

$$p = -\rho g\left(\frac{a}{g}x + z\right)$$

Free surface equation:

$$ax + gz = 0$$

$$\mathrm{tg}\alpha = -\frac{z}{x} = \frac{a}{g} = \frac{0.98}{9.8} = 0.1$$

$$\alpha = 5°45'$$

Pressure at B:

$$p_B = -\rho g\left(\frac{a}{g}x + z\right)$$

$$= -9800\left(\frac{0\cdot 98}{9\cdot 8}\times(-1.5) + (-1.0)\right)$$

$$= 11270\,\mathrm{N/m^2} = 11.27\,\mathrm{kPa}$$

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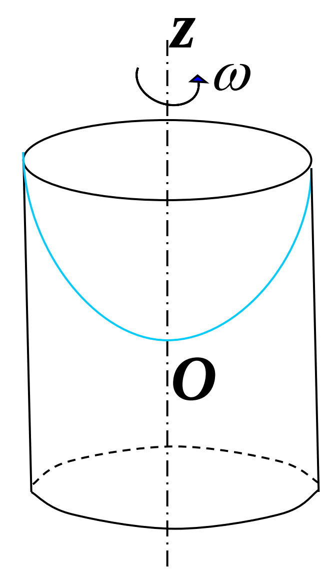
Questions

1,What is an isobaric surface? How do you determine if a surface is isobaric?

Points of an isobaric surface have the same pressure. In a static, homogeneous, connected fluid, points at the same level have the same pressure.

2, Is the isobaric surface of a fluid at relative equilibrium a horizontal surface? Why? Under what conditions is an isobaric plane a horizontal plane?



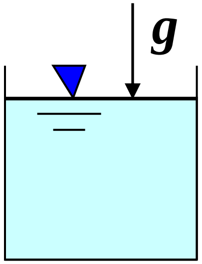
Not necessarily. It is a horizontal plane only if the net mass force (gravity, inertia) is in the vertical direction.

3, Is the pressure measured on the pressure gauge the absolute pressure or relative pressure?

Relative pressure.

4. When an open container filled with liquid is in free fall, what is the pressure distribution on the wall of the container?

$$\because \mathrm{d}p = \rho(f_x \mathrm{d}x + f_y \mathrm{d}y + f_z \mathrm{d}z) = \rho(g - g)\mathrm{d}z = 0$$



∴ ***p* = const** Free surface: ***p* = 0**

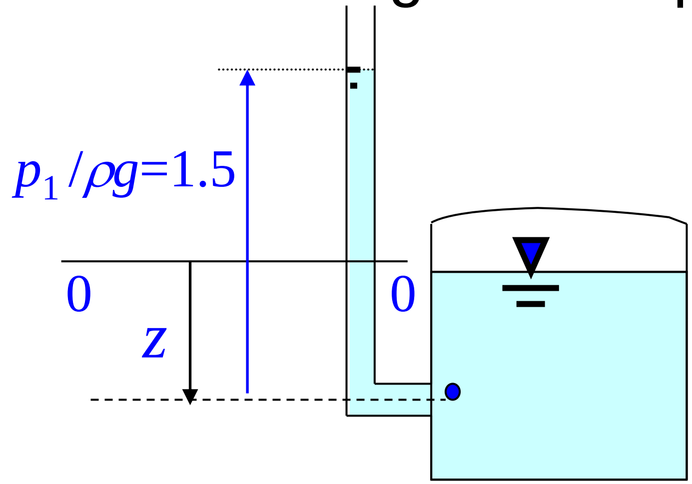
∴ ***p* = 0**

5, If the maximum pressure a person can withstand is 1.274 MPa (relative pressure), what is the diver's limit diving depth?

$$h = \frac{p}{\rho g} = \frac{1.274 \times 10^6}{9800} = 130\ \mathrm{m}$$

6, If the hydraulic head is **1m** and the pressure head is **1.5m**, what should be the minimum height of the piezometer tube?

Answer: **1.5m**



7, Why can a siphon transport water to a certain height?

Because of the vacuum inside the siphon.

8. In a static fluid, are the hydraulic heads at each point equal? How about a flowing fluid?

In a static fluid, yes. In a flowing fluid, generally not equal.